

Rukhshan Haroon

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EDUCATION

Tufts University, Greater Boston, MA, USA.

Ph.D. in Computer Science, Sept. 2022 - present

Specialization: Accessibility, Human-Computer Interaction (HCI), Artificial Intelligence

Lahore University of Management Sciences (LUMS), Lahore, Pakistan.

B.Sc. in Computer Science, Sept. 2018 - May 2022

SKILLS

Research: Semi-Structured Interviews, Think-Aloud Method, User Study & Experimental Design, Quantitative Methods, Survey Design, AI Bias & Performance Evaluation, Large-Scale System Design.

Technical: Agentic AI, Full-Stack Development, Python, PyTorch, C/C++, Unity, MongoDB, Git.

Selected Coursework: Assistive Design (Mechanical Eng.), Generative AI for Social Impact, HCI for Disability, Deep Learning, Statistics, Advanced Programming, Data Science, Data Structures.

WORK EXPERIENCE

Research Lead (AI, Accessibility, and HCI)—Tufts University, Boston, MA. *Sept 2022 - present.*

- Designed and prototyped an LLM-based: (1) training platform that uses conversational simulations to help non-autistic users improve cross-neurotype conversational skills; (2) writing assistant to support autistic users with self-expression and interpretation of linguistic nuances.
- Designed and led user studies with **30+** autistic users to evaluate the aforementioned systems.
- Conducted an empirical study of LLMs on tasks involving evaluation of cross-neurotype communication differences; contributed a comprehensive LLM evaluation framework and dataset.

Researcher (AI, Systems, and HCI)—Tufts University, Boston, MA. *Jan - Dec 2024.*

- Designed a cost-saving proxy for LLM routing, reducing costs by up to **50%** without quality loss.
- Deployed the proxy in **3 countries** over **12+ months** supporting **85k+ user-requests**.

Research Lead (Security, Privacy, and HCI)—LUMS, Lahore, Pakistan. *May 2020 - Sept 2022.*

- Led mixed-methods research in Pakistan to study: (1) gender disparities among healthcare workers in COVID-19; (2) how socio-cultural factors shaped pandemic-related misinformation spread.
- Implemented a benchmarking framework to compare CVEs, size, correctness, and build time of software debloating tools; built multi-tool pipelines achieving **70%** size and **40%** CVE reductions.

SELECTED RESEARCH

ASSETS is the flagship ACM SIGACCESS conference and the premier venue on accessible HCI.

Rukhshan Haroon, Kyle Wigdor, Katie Yang, Nicole Toumanios, Eileen Crehan, and Fahad Dogar. 2025. *NeuroBridge: Using Generative AI to Bridge Cross-neurotype Communication Differences through Neurotypical Perspective-taking*. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*. 🏆 **Best Student Paper** [PDF](#)

Rukhshan Haroon and Fahad Dogar. 2024. *TwIPS: A Large Language Model Powered Texting Application to Simplify Conversational Nuances for Autistic Users*. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24)*. [PDF](#)

Rukhshan Haroon, Haisum Haroon, Wren Kritzer, Jaylin Herskovitz, and Fahad Dogar. 2026. *Bridging or Widening the Gap? An Empirical Study on LLM Bias Against Autistic Communication Styles*. Under Submission at ACM ASSETS 2026. [PDF](#)

Rukhshan Haroon. 2026. *Towards AI-Mediated Bidirectional Communication Support for Cross-Neurotype Interactions*. SIGACCESS Accessibility and Computing. [PDF](#)

Rukhshan Haroon, Ayesha Naeem, Priya Sajjad, and Zartash Uzmi. 2024. *On the Frontline During the COVID-19 Pandemic: Gender Inequality and Experiences of Healthcare Workers in Pakistan*. *ACM Journal on Computing and Sustainable Societies*. [PDF](#)

SERVICE AND VOLUNTEERING

Reviewer, ACM CHI, ACM IUI, ACM TACCESS, and ACM UIST 2025

HONORS AND AWARDS

Best Student Paper, ACM ASSETS, [Press](#) 2025

NSF Travel Award, ACM ASSETS Doctoral Consortium 2025

Augmented Reality Hackathon Winner, Harvard University, [Blog](#) 2023

Award of Academic Distinction, LUMS 2020–2022

Students as Co-Researchers Grant, LUMS 2021